davebulaval Fix typo in title (#614) 1d150f0 · 4 years ago

214 lines (171 loc) · 28.7 KB

Summarization

Summarization is the task of producing a shorter version of one or several documents that preserves most of the input's meaning.

Warning: Evaluation Metrics

For summarization, automatic metrics such as ROUGE and METEOR have serious limitations:

1. They only assess content selection and do not account for other quality aspects, such as fluency, grammaticality, coherence, etc.
2. To assess content selection, they rely mostly on lexical overlap, although an abstractive summary could express they same content as a reference without any lexical overlap.
3. Given the subjectiveness of summarization and the correspondingly low agreement between annotators, the metrics were designed to be used with multiple reference summaries per input. However, recent datasets such as CNN/DailyMail and Gigaword provide only a single reference.

Therefore, tracking progress and claiming state-of-the-art based only on these metrics is questionable. Most papers carry out additional manual comparisons of alternative summaries. Unfortunately, such experiments are difficult to compare across papers. If you have an idea on how to do that, feel free to contribute.

CNN / Daily Mail

The CNN / Daily Mail dataset as processed by Nallapati et al. (2016) has been used for evaluating summarization. The dataset contains online news articles (781 tokens on average) paired with multi-sentence summaries (3.75 sentences or 56 tokens on average). The processed version contains 287,226 training pairs, 13,368 validation pairs and 11,490 test pairs. Models are evaluated with full-length F1-scores of ROUGE-1, ROUGE-2, ROUGE-L, and METEOR (optional). The multilingual version of CNN / Daily Mail dataset exists and is available for five different languages (French, German, Spanish, Russian, Turkish).

Anonymized version

The following models have been evaluated on the entitiy-anonymized version of the dataset introduced by Nallapati et al. (2016).

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source
RNES w/o coherence (Wu and Hu, 2018)	41.25	18.87	37.75	-	Learning to Extract Coherent Summary via Deep Reinforcement Learning
SWAP-NET (Jadhav and Rajan, 2018)	41.6	18.3	37.7	-	Extractive Summarization with SWAP-NET: Sentences and Words from Alternating Pointer Networks
HSASS (Al-Sabahi et al., 2018)	42.3	17.8	37.6	-	A Hierarchical Structured Self-Attentive Model for Extractive Document Summarization (HSSAS)

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source
GAN (Liu et al., 2018)	39.92	17.65	36.71	-	Generative Adversarial Network for Abstractive Text Summarization
KIGN+Prediction-guide (Li et al., 2018)	38.95	17.12	35.68	-	Guiding Generation for Abstractive Text Summarization based on Key Information Guide Network
SummaRuNNer (Nallapati et al., 2017)	39.6	16.2	35.3	-	SummaRuNNer: A Recurrent Neural Network based Sequence Model for Extractive Summarization of Documents
rnn-ext + abs + RL + rerank (Chen and Bansal, 2018)	39.66	15.85	37.34	-	Fast Abstractive Summarization with Reinforce-Selected Sentence Rewriting
ML+RL, with intra-attention (Paulus et al., 2018)	39.87	15.82	36.90	-	A Deep Reinforced Model for Abstractive Summarization
Lead-3 baseline (Nallapati et al., 2017)	39.2	15.7	35.5	-	SummaRuNNer: A Recurrent Neural Network

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source
					based Sequence Model for Extractive Summarization of Documents
ML+RL ROUGE+Novel, with LM (Kryscinski et al., 2018)	40.02	15.53	37.44	-	Improving Abstraction in Text Summarization
(Tan et al., 2017)	38.1	13.9	34.0	-	Abstractive Document Summarization with a Graph-Based Attentional Neural Model
words-lvt2k-temp-att (Nallapti et al., 2016)	35.46	13.30	32.65	-	Abstractive Text Summarization using Sequence-to-sequence RNNs and Beyond

Non-Anonymized Version: Extractive Models

The following models have been evaluated on the non-anonymized version of the dataset introduced by See et al. (2017).

The first table covers Extractive Models, while the second covers abstractive approaches.

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source	Code
MatchSum (Zhong et al., 2020)	44.41	20.86	40.55	-	Extractive Summarization as Text Matching	Offici
DiscoBERT w.G_R & G_C (Xu et al. 2019)	43.77	20.85	40.67	-	A Discourse-Aware Neural Extractive Model for Text Summarization	Offici
BertSumExt (Liu and Lapata 2019)	43.85	20.34	39.90	-	Text Summarization with Pretrained Encoders	Offici
BERT-ext + RL (Bae et al., 2019)	42.76	19.87	39.11	-	Summary Level Training of Sentence Rewriting for Abstractive Summarization	
PNBERT (Zhong et al., 2019)	42.69	19.60	38.85	-	Searching for Effective Neural Extractive Summarization: What Works and What's Next	Offici
HIBERT (Zhang et al., 2019)	42.37	19.95	38.83	-	HIBERT: Document Level Pre-training of Hierarchical Bidirectional Transformers	

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source	Code
					for Document Summarization	
NeuSUM (Zhou et al., 2018)	41.59	19.01	37.98	-	Neural Document Summarization by Jointly Learning to Score and Select Sentences	Offici
Latent (Zhang et al., 2018)	41.05	18.77	37.54	-	Neural Latent Extractive Document Summarization	
BanditSum (Dong et al., 2018)	41.5	18.7	37.6	-	BANDITSUM: Extractive Summarization as a Contextual Bandit	Offici
REFRESH (Narayan et al., 2018)	40.0	18.2	36.6	-	Ranking Sentences for Extractive Summarization with Reinforcement Learning	Offici
Lead-3 baseline (See et al., 2017)	40.34	17.70	36.57	22.21	Get To The Point: Summarization with Pointer-Generator Networks	Offici

Non-Anonymized: Abstractive Models & Mixed Models

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source	
BRIO (Liu et al., 2022)	47.78	23.55	44.57	-	BRIO: Bringing Order to Abstractive Summarization	C
SimCLS (Liu et al., 2021)	46.67	22.15	43.54	-	SimCLS: A Simple Framework for Contrastive Learning of Abstractive Summarization	C
					GSum: A General	

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ProphetNet (Yan, Qi, Gong, Liu et al., 2020)	44.20	21.17	41.30	-	ProphetNet: Predicting Future N-gram for Sequence-to-Sequence Pre-training	C
PEGASUS (Zhang et al., 2019)	44.17	21.47	41.11	-	PEGASUS: Pre-training with Extracted Gap-sentences for Abstractive Summarization	C
BART (Lewis et al., 2019)	44.16	21.28	40.90	-	BART: Denoising Sequence-to-Sequence Pre-training for Natural	C

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source	
					Language Generation, Translation, and Comprehension	
T5 (Raffel et al., 2019)	43.52	21.55	40.69	-	Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer	C
UniLM (Dong et al., 2019)	43.33	20.21	40.51	-	Unified Language Model Pre-training for Natural Language Understanding and Generation	C
CNN-2sent-hieco-RBM (Zhang et al., 2019)	42.04	19.77	39.42	-	Abstract Text Summarization with a Convolutional Seq2Seq Model	
BertSumExtAbs (Liu and Lapata 2019)	42.13	19.60	39.18	-	Text Summarization with Pretrained Encoders	C
BERT-ext + abs + RL + rerank (Bae et al., 2019)	41.90	19.08	39.64	-	Summary Level Training of Sentence Rewriting for Abstractive Summarization	

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source	
Two-Stage + RL (Zhang et al., 2019)	41.71	19.49	38.79	-	Pretraining-Based Natural Language Generation for Text Summarization	
DCA (Celikyilmaz et al., 2018)	41.69	19.47	37.92	-	Deep Communicating Agents for Abstractive Summarization	
EditNet (Moroshko et al., 2018)	41.42	19.03	38.36	-	An Editorial Network for Enhanced Document Summarization	
rnn-ext + RL (Chen and Bansal, 2018)	41.47	18.72	37.76	22.35	Fast Abstractive Summarization with Reinforce-Selected Sentence Rewriting	C
Bottom-Up Summarization (Gehrmann et al., 2018)	41.22	18.68	38.34	-	Bottom-Up Abstractive Summarization	C
(Li et al., 2018a)	41.54	18.18	36.47	-	Improving Neural Abstractive Document Summarization with Explicit Information Selection Modeling	

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source	
(Li et al., 2018b)	40.30	18.02	37.36	-	Improving Neural Abstractive Document Summarization with Structural Regularization	
ROUGESal+Ent RL (Pasunuru and Bansal, 2018)	40.43	18.00	37.10	20.02	Multi-Reward Reinforced Summarization with Saliency and Entailment	
end2end w/ inconsistency loss (Hsu et al., 2018)	40.68	17.97	37.13	-	A Unified Model for Extractive and Abstractive Summarization using Inconsistency Loss	
RL + pg + cbdec (Jiang and Bansal, 2018)	40.66	17.87	37.06	20.51	Closed-Book Training to Improve Summarization Encoder Memory	
rnn-ext + abs + RL + rerank (Chen and Bansal, 2018)	40.88	17.80	38.54	20.38	Fast Abstractive Summarization with Reinforce-Selected Sentence Rewriting	C
Pointer + Coverage + EntailmentGen	39.81	17.64	36.54	18.54	Soft Layer-Specific Multi-Task	

Model	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Paper / Source	C
+ QuestionGen (Guo et al., 2018)					Summarization with Entailment and Question Generation	
ML+RL ROUGE+Novel, with LM (Kryscinski et al., 2018)	40.19	17.38	37.52	-	Improving Abstraction in Text Summarization	
Pointer-generator + coverage (See et al., 2017)	39.53	17.28	36.38	18.72	Get To The Point: Summarization with Pointer-Generator Networks	C

Gigaword

The Gigaword summarization dataset has been first used by Rush et al., 2015 and represents a sentence summarization / headline generation task with very short input documents (31.4 tokens) and summaries (8.3 tokens). It contains 3.8M training, 189k development and 1951 test instances. Models are evaluated with ROUGE-1, ROUGE-2 and ROUGE-L using full-length F1-scores.

Below Results are ranking by ROUGE-2 Scores.

Model	ROUGE-1	ROUGE-2*	ROUGE-L	Paper / Source	Code
ControlCopying (Song et al., 2020)	39.08	20.47	36.69	Controlling the Amount of Verbatim Copying in Abstractive Summarization	Official

Model	ROUGE-1	ROUGE-2*	ROUGE-L	Paper / Source	Code
ProphetNet (Yan, Qi, Gong, Liu et al., 2020)	39.51	20.42	36.69	ProphetNet: Predicting Future N-gram for Sequence-to-Sequence Pre-training	Official
UniLM (Dong et al., 2019)	38.90	20.05	36.00	Unified Language Model Pre-training for Natural Language Understanding and Generation	Official
PEGASUS (Zhang et al., 2019)	39.12	19.86	36.24	PEGASUS: Pre-training with Extracted Gap-sentences for Abstractive Summarization	Official
BiSET (Wang et al., 2019)	39.11	19.78	36.87	BiSET: Bi-directional Selective Encoding with Template for Abstractive Summarization	Official
MASS (Song et al., 2019)	38.73	19.71	35.96	MASS: Masked Sequence to Sequence Pre-training for Language Generation	Official
Re^3 Sum (Cao et al., 2018)	37.04	19.03	34.46	Retrieve, Rerank and	

Model	ROUGE-1	ROUGE-2*	ROUGE-L	Paper / Source	Code
				Rewrite: Soft Template Based Neural Summarization	
JointParsing (Song el at., 2020)	36.61	18.85	34.33	Joint Parsing and Generation for Abstractive Summarization	Official
CNN-2sent-hieco-RBM (Zhang et al., 2019)	37.95	18.64	35.11	Abstract Text Summarization with a Convolutional Seq2Seq Model	
Reinforced-Topic-ConvS2S (Wang et al., 2018)	36.92	18.29	34.58	A Reinforced Topic-Aware Convolutional Sequence-to-Sequence Model for Abstractive Text Summarization	
CGU (Lin et al., 2018)	36.3	18.0	33.8	Global Encoding for Abstractive Summarization	Official
Pointer + Coverage + EntailmentGen + QuestionGen (Guo et al., 2018)	35.98	17.76	33.63	Soft Layer-Specific Multi-Task Summarization with Entailment and Question Generation	

Model	ROUGE-1	ROUGE-2*	ROUGE-L	Paper / Source	Code
Struct+2Way+Word (Song et al., 2018)	35.47	17.66	33.52	Structure-Infused Copy Mechanisms for Abstractive Summarization	Official
FTSum_g (Cao et al., 2018)	37.27	17.65	34.24	Faithful to the Original: Fact Aware Neural Abstractive Summarization	
DRGD (Li et al., 2017)	36.27	17.57	33.62	Deep Recurrent Generative Decoder for Abstractive Text Summarization	
SEASS (Zhou et al., 2017)	36.15	17.54	33.63	Selective Encoding for Abstractive Sentence Summarization	Official
EndDec+WFE (Suzuki and Nagata, 2017)	36.30	17.31	33.88	Cutting-off Redundant Repeating Generations for Neural Abstractive Summarization	
Seq2seq + selective + MTL + ERAM (Li et al., 2018)	35.33	17.27	33.19	Ensure the Correctness of the Summary: Incorporate Entailment Knowledge into Abstractive	

Model	ROUGE-1	ROUGE-2*	ROUGE-L	Paper / Source	Code
				Sentence Summarization	
Seq2seq + E2T_cnn (Amplayo et al., 2018)	37.04	16.66	34.93	Entity Commonsense Representation for Neural Abstractive Summarization	
RAS-Elman (Chopra et al., 2016)	33.78	15.97	31.15	Abstractive Sentence Summarization with Attentive Recurrent Neural Networks	
words-lvt5k-1sent (Nallapti et al., 2016)	32.67	15.59	30.64	Abstractive Text Summarization using Sequence-to-sequence RNNs and Beyond	
ABS+ (Rush et al., 2015)	29.76	11.88	26.96	A Neural Attention Model for Sentence Summarization *	
ABS (Rush et al., 2015)	29.55	11.32	26.42	A Neural Attention Model for Sentence Summarization *	

(*) Rush et al., 2015 report ROUGE recall, the table here contains ROUGE F1-scores for Rush's model reported by Chopra et al., 2016

X-Sum

X-Sum (standing for *Extreme Summarization*), introduced by Narayan et al., 2018, is a summarization dataset which does not favor extractive strategies and calls for an abstractive modeling approach.
The idea of this dataset is to create a short, one sentence news summary.
Data is collected by harvesting online articles from the BBC.
The dataset contain **204 045** samples for the training set, **11 332** for the validation set, and **11 334** for the test set. In average the length of article is 431 words (~20 sentences) and the length of summary is 23 words. It can be downloaded here.
Evaluation metrics are ROUGE-1, ROUGE-2 and ROUGE-L.

Model	ROUGE-1	ROUGE-2	ROUGE-L	Paper / Source	Code
BRIO (Liu et al., 2022)	49.07	25.59	40.40	BRIO: Bringing Order to Abstractive Summarization	Official
PEGASUS (Zhang et al., 2019)	47.21	24.56	39.25	PEGASUS: Pre-training with Extracted Gap-sentences for Abstractive Summarization	Official
BART (Lewis et al., 2019)	45.14	22.27	37.25	BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension	Official
BertSumExtAbs (Liu et al., 2019)	38.81	16.50	31.27	Text Summarization with Pretrained Encoders	Official

Model	ROUGE-1	ROUGE-2	ROUGE-L	Paper / Source	Code
T-ConvS2S	31.89	11.54	25.75	Don't Give Me the Details, Just the Summary!	Official
PtGen	29.70	9.21	23.24	Don't Give Me the Details, Just the Summary!	Official
Seq2Seq	28.42	8.77	22.48	Don't Give Me the Details, Just the Summary!	Official
PtGen-Covg	28.10	8.02	21.72	Don't Give Me the Details, Just the Summary!	Official
Baseline : Extractive Oracle	29.79	8.81	22.66	Don't Give Me the Details, Just the Summary!	Official
Baseline : Lead-3	16.30	1.60	11.95	Don't Give Me the Details, Just the Summary!	Official
Baseline : Random	15.16	1.78	11.27	Don't Give Me the Details, Just the Summary!	Official

DUC 2004 Task 1

Similar to Gigaword, task 1 of DUC 2004 is a sentence summarization task. The dataset contains 500 documents with on average 35.6 tokens and summaries with 10.4 tokens. Due to its size, neural models are typically trained on other datasets and only tested on DUC 2004. Evaluation metrics are ROUGE-1, ROUGE-2 and ROUGE-L recall @ 75 bytes.

Model	ROUGE-1	ROUGE-2	ROUGE-L	Paper / Source	Code
Transformer + LRPE + PE + ALONE + RE-	32.57	11.63	28.24	All Word Embeddings from One Embedding	Official

Model	ROUGE-1	ROUGE-2	ROUGE-L	Paper / Source	Code
ranking (Takase and Kobayashi, 2020)					
Transformer + LRPE + PE + Re-ranking (Takase and Okazaki, 2019)	32.29	11.49	28.03	Positional Encoding to Control Output Sequence Length	Official
DRGD (Li et al., 2017)	31.79	10.75	27.48	Deep Recurrent Generative Decoder for Abstractive Text Summarization	
EndDec+WFE (Suzuki and Nagata, 2017)	32.28	10.54	27.8	Cutting-off Redundant Repeating Generations for Neural Abstractive Summarization	
Reinforced-Topic-ConvS2S (Wang et al., 2018)	31.15	10.85	27.68	A Reinforced Topic-Aware Convolutional Sequence-to-Sequence Model for Abstractive Text Summarization	
CNN-2sent-hieco-RBM (Zhang et al., 2019)	29.74	9.85	25.81	Abstract Text Summarization with a Convolutional Seq2Seq Model	

Model	ROUGE-1	ROUGE-2	ROUGE-L	Paper / Source	Code
Seq2seq + selective + MTL + ERAM (Li et al., 2018)	29.33	10.24	25.24	Ensure the Correctness of the Summary: Incorporate Entailment Knowledge into Abstractive Sentence Summarization	
SEASS (Zhou et al., 2017)	29.21	9.56	25.51	Selective Encoding for Abstractive Sentence Summarization	
words-lvt5k-1sent (Nallapti et al., 2016)	28.61	9.42	25.24	Abstractive Text Summarization using Sequence-to-sequence RNNs and Beyond	
ABS+ (Rush et al., 2015)	28.18	8.49	23.81	A Neural Attention Model for Sentence Summarization	
RAS-Elman (Chopra et al., 2016)	28.97	8.26	24.06	Abstractive Sentence Summarization with Attentive Recurrent Neural Networks	
ABS (Rush et al., 2015)	26.55	7.06	22.05	A Neural Attention Model for Sentence Summarization	

Webis-TLDR-17 Corpus

This dataset contains 3 Million pairs of content and self-written summaries mined from Reddit. It is one of the first large-scale summarization dataset from the social media domain. For more details, refer to TL;DR: Mining Reddit to Learn Automatic Summarization

Model	ROUGE-1	ROUGE-2	ROUGE-L	Paper/Source	Code
Transformer + Copy (Gehrmann et al., 2019)	22	6	17	Generating Summaries with Finetuned Language Models	
Unified VAE + PGN (Choi et al., 2019)	19	4	15	VAE-PGN based Abstractive Model in Multi-stage Architecture for Text Summarization	

Webis-Snippet-20 Corpus

This dataset contains approximately 10 Million (webpage content, abstractive snippet) pairs and 3.5 Million (query term, webpage content, abstractive snippet) triples for the novel task of (query-biased) abstractive snippet generation of web pages. The corpus is compiled from ClueWeb09, ClueWeb12 and the DMOZ Open Directory Project. For more details, refer to Abstractive Snippet Generation

Model	ROUGE-1	ROUGE-2	ROUGE-L	Usefulness	Paper/Source	Code
Anchor-context + Query biased (Chen et al., 2020)	25.7	5.2	20.1	66.18	Abstractive Snippet Generation	

Sentence Compression

Sentence compression produces a shorter sentence by removing redundant information, preserving the grammatically and the important content of the original sentence.

Google Dataset

The Google Dataset was built by Filippova et al., 2013(Overcoming the Lack of Parallel Data in Sentence Compression). The first dataset released contained only 10,000 sentence-compression pairs, but last year was released an additional 200,000 pairs.

Example of a sentence-compression pair:

> Sentence: Floyd Mayweather is open to fighting Amir Khan in the future, despite snubbing the Bolton-born boxer in favour of a May bout with Argentine Marcos Maidana, according to promoters Golden Boy

> Compression: Floyd Mayweather is open to fighting Amir Khan in the future.

In short, this is a deletion-based task where the compression is a subsequence from the original sentence. From the 10,000 pairs of the eval portion(repository) it is used the very first 1,000 sentence for automatic evaluation and the 200,000 pairs for training.

Models are evaluated using the following metrics:

- F1 - compute the recall and precision in terms of tokens kept in the golden and the generated compressions.
- Compression rate (CR) - the length of the compression in characters divided over the sentence length.

Model	F1	CR	Paper / Source	Code
SLAHAN with syntactic information (Kamigaito et al. 2020)	0.855	0.407	Syntactically Look-Ahead Attention Network for Sentence Compression	https://github.com/kamigaito/SLAHAN
BiRNN + LM Evaluator	0.851	0.39	A Language Model based Evaluator for	https://github.com/code4conference/c

Model	F1	CR	Paper / Source	Code
(Zhao et al. 2018)			Sentence Compression	
LSTM (Filippova et al., 2015)	0.82	0.38	Sentence Compression by Deletion with LSTMs	
BiLSTM (Wang et al., 2017)	0.8	0.43	Can Syntax Help? Improving an LSTM-based Sentence Compression Model for New Domains	

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